

IAN SAMPLE QUESTIONS

3 MARKS

1. What are some key Python libraries used for AI and machine learning, and how do they support AI development? / Give an account on the commonly used python packages for natural language processing and their functionality. / Elaborate on a widely used Python package for NLP. List 5 tasks it supports.
2. What is the regex pattern that can validate dates in DD/MM/YYYY format:
3. How can we calculate the similarity between two vectors? Explain with an example.
4. How can text rank be computed for a given document/documents? Illustrate its role in implementing text summarization.
5. What is the significance of language modeling in natural language processing

7 MARKS

1. Discuss in detail the applications of NLP.
2. Compare the different types of commonly employed tokenization methods.
3. Compute TF-IDF for the given sentences
S1: I was driving my car to the college.
S2: The driver of the car left from college.
S3: College buses were not running due to strike.

OR

Compute TF-IDF score for the given sentences

S1: "The cat sat on the mat."

S2: "The dog sat on the mat."

S3: "The cat chased the dog."

OR

Compute TF-IDF score for the given sentences

S1: faster than the wind but slower than light

S2: lighter than feather but heavier than air.

S3: higher than sun but lower than stars.

OR

Compute TF-IDF for the given sentences

S1: lower the speed, tastier the food becomes.

S2: tastier the food is, more you want to eat.

S3: more you eat, faster the speed.

4. Explain Latent Dirichlet Allocation